TAMIL NADU STATE BOARD - STANDARD 11 CHEMISTRY VOLUME 1

CHAPTER 7: THERMODYNAMICS

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Chemistry Volume 1
Chapter Name:	Thermodynamics

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

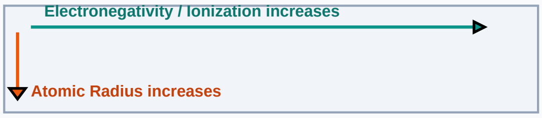
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Enthalpy	The total heat content of a system, equal to internal energy plus product of pressure and volume (H = U + PV).
Hess Law of Constant Heat Summation	The enthalpy change of a chemical reaction is constant, whether it occurs in one step or in several steps.
Gibbs Free Energy	A thermodynamic potential that measures the maximum reversible work performable by a system, predicting reaction spontaneity.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

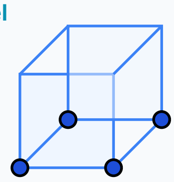
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Simple Cubic Lattice System

Essential Formulas, Laws & Identities:

- First Law: delta_U = q + w | Expansion Work: w = -P * delta_V
- Gibbs Relation: delta_G = delta_H - T * delta_S
- Spontaneity: delta_G < 0 (spontaneous), delta_G = 0 (equilibrium), delta_G > 0 (non-spontaneous)

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

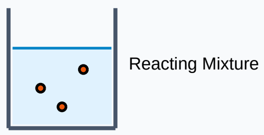
Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Calculate Gibbs free energy change for a reaction if delta_H = -110 kJ, delta_S = +150 J/K at 298K.

Step-by-step Solution:
delta_G = delta_H - T*delta_S.
Convert delta_S to kJ: 150/1000 = 0.15 kJ/K.
delta_G = -110 - 298*(0.15) = -110 - 44.7 = -154.7 kJ. (Spontaneous)

Titration Beaker & Solute Precipitation



HOTS (Higher Order Thinking Skills) Challenge

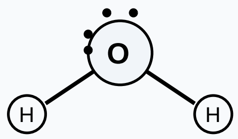
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Derive the relation delta_H = delta_U + delta_n_g * R * T for gaseous chemical reactions.

Analytical Solution Strategy:

By definition, H = U + PV => delta_H = delta_U + delta_(PV).
For ideal gases, PV = nRT. At constant temperature: delta_(PV) = delta_(nRT) = delta_n_g * R * T.
Substitute to get: delta_H = delta_U + delta_n_g * R * T.

Lewis Dot Structure (H_2O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Distinguish between state functions and path functions.

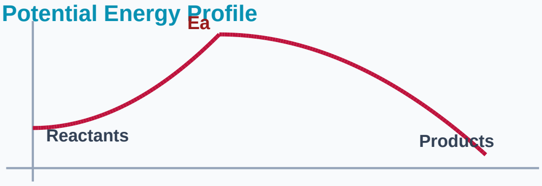
Q2 (2-Mark): State the First and Third Laws of Thermodynamics.

Q3 (2-Mark): State Hess's Law and list two applications.

Q4 (5-Mark): What is entropy? Describe its changes when liquid freezes.

Q5 (5-Mark): Define standard enthalpy of formation.

Chemical Reaction Potential Energy Profile



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Chemical Engineering

Reaction spontaneity is calculated using Gibbs values to avoid explosive mixtures.

Application Case: Internal Combustion

Engine design efficiency is optimized by mapping enthalpy releases of fuel combustion.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C_6H_6) Resonance

Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

Electrochemical Galvanic Cell System